

# The GNoME Low- $\kappa_L$ Screen

Architecture, physics, and the crystallographic structure of 33,323 candidates

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## Abstract

A three-model CGCNN ensemble predicts bulk and shear modulus for every material in a 554,054-entry snapshot of Google DeepMind’s GNoME discovery database; the paper’s own Slack-model physics turns those moduli into a predicted lattice thermal conductivity  $\kappa_L$ ; and 16,299 candidates clear the screening threshold of  $\kappa_L \leq 1 \text{ W m}^{-1} \text{ K}^{-1}$ . That screen, and the model behind it, are described in full in `docs/method.tex`. This document is the follow-on: having a candidate list is not the same as understanding it, so every one of those 33,323 scored candidates is here tagged with its crystallographic space group (read directly from GNoME’s own structure metadata) and its constituent elements, then broken down two ways — across the full screen, and across the 11,612 oxide-containing candidates specifically — to ask what the low- $\kappa_L$  region of chemical space actually looks like. Two findings stand out. First, low- $\kappa_L$  candidates are not uniformly distributed across chemistry: halogens (Br, I, Cl, F) are markedly *enriched* among low- $\kappa_L$  hits (up to  $1.9\times$  their background rate) while oxygen is markedly *depleted* ( $0.27\times$ ), consistent with soft, heavy-halide lattices having lower sound velocities and Debye temperatures than tightly-bonded oxides. Second, both the general screen and the oxide subset skew heavily toward low-symmetry space groups (triclinic and monoclinic account for two-thirds of all candidates), a pattern this document quantifies precisely rather than leaves as an impression. Finally, we identify the single strongest candidate in every crystal system and every space group, for both populations — ranked not by the raw point-estimate  $\kappa_L$ , but by its 95th-percentile Monte Carlo value, since ranking  $> 30\text{k}$  candidates by the point estimate alone turns out to be a textbook winner’s-curse setup: the top point-estimate oxide and the runner-up general candidate both carry  $19\text{--}22\times$  p95/p05 uncertainty ratios, indistinguishable from noise rather than a confident low- $\kappa_L$  prediction. Re-ranked this way, **CrSi<sub>3</sub>Cl<sub>16</sub>** (triclinic,  $P1$ ) holds its #1 spot in the general screen (p95 =  $0.0094 \text{ W m}^{-1} \text{ K}^{-1}$ ,  $7.6\times$  ratio); the oxide leaderboard reorders, with **COF<sub>14</sub>** (monoclinic,  $P2_1$ ) taking the top spot at a much tighter  $2.3\times$  ratio. **That p95/p05 ranking itself turned out to be overconfident:** checked against 1,648 real DFT-labelled test crystals (Section 6), the ensemble’s claimed 90% interval actually covers only 48–51% of true values — a validated, out-of-sample recalibration widens every interval  $4.3\text{--}4.5\times$ , after which COF<sub>14</sub> still holds its #1 oxide spot, but CrSi<sub>3</sub>Cl<sub>16</sub> is displaced from the general-screen top spot by **SbP<sub>2</sub>Cl<sub>13</sub>**. The ranking survives better than any individual confidence claim does.

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## 1 Background and pipeline

The full pipeline, trained model, and validation are described in `docs/method.tex`; this section recaps only what is needed to read the results below.



Every material in this document already passed the paper’s own three-stage funnel (bandgap, thermodynamic stability, no radioactive elements) before being scored:

Table 1: The screening funnel, this project’s current 554,054-material GNoME snapshot.

| Stage                                                           | Candidates remaining                     |
|-----------------------------------------------------------------|------------------------------------------|
| GNoME snapshot                                                  | 554,054                                  |
| Bandgap [0.1, 3.0] eV, decomposition energy $\leq 0$            | 37,889                                   |
| + no radioactive elements                                       | <b>33,323</b> (scored by CGCNN)          |
| + $\kappa_L \leq 1 \text{ W m}^{-1} \text{ K}^{-1}$ (own model) | <b>16,299</b> (“low- $\kappa_L$ ” below) |

## 2 Model and physics summary

### 2.1 The network: CGCNN

Each crystal becomes a graph — atoms as nodes, bonds as edges — and the network predicts  $K_{\text{VRH}}$  and  $G_{\text{VRH}}$  (Voigt–Reuss–Hill bulk and shear modulus) from it directly, with no DFT calculation involved.

- **Nodes.** Each atom is a 92-dimensional vector of tabulated element properties (group, period, electronegativity, radius, . . . ), looked up by atomic number, not learned.
- **Edges.** Every atom’s 12 nearest neighbours within 8 Å, respecting periodic boundary conditions, with the bond distance expanded onto 41 Gaussian basis functions rather than used as a single raw number.
- **Convolution (gated message passing), three layers.** For atom  $i$  with neighbours  $j$  and bond feature  $\mathbf{e}_{ij}$ :

$$\mathbf{v}_i^{(t+1)} = \mathbf{v}_i^{(t)} + \sum_j \underbrace{\sigma(\mathbf{z}_{ij}^{(t)} \mathbf{W}_f + \mathbf{b}_f)}_{\text{gate}} \odot \underbrace{g(\mathbf{z}_{ij}^{(t)} \mathbf{W}_s + \mathbf{b}_s)}_{\text{message}}, \quad \mathbf{z}_{ij}^{(t)} = \mathbf{v}_i^{(t)} \oplus \mathbf{v}_j^{(t)} \oplus \mathbf{e}_{ij} \quad (1)$$

with  $\sigma$  a sigmoid (how much of a neighbour’s message to admit) and  $g$  a softplus (what that message says). Shared weights across every edge are what make this a *convolution*; the residual term lets gradients reach every layer.

- **Pooling.** The crystal-level vector is the *mean* over its atoms (not the sum), so the prediction does not change if the unit cell is doubled — essential for an intensive property like a modulus.
- **Ensembling.** Three independently-initialised models per target, averaged in log space; their spread is a free per-crystal uncertainty estimate with no equivalent in a single pre-trained model.

## 2.2 The physics: Slack model

Everything from  $(K, G)$  to  $\kappa_L$  is closed-form physics, unchanged from the paper, driven by our own predicted moduli:

$$v_{\text{long}} = \sqrt{(K + \frac{4}{3}G)/\rho}, \quad v_{\text{trans}} = \sqrt{G/\rho} \quad (2)$$

$$\gamma = \frac{3(1 + \nu)}{2(2 - 3\nu)}, \quad \nu = \frac{(v_{\text{long}}/v_{\text{trans}})^2 - 2}{2(v_{\text{long}}/v_{\text{trans}})^2 - 2} \quad (3)$$

$$\kappa_L = \frac{G \cdot v_{\text{sound}} \cdot V^{1/3}}{N_{\text{atoms}} \cdot 300} e^{-\gamma} \quad (\text{Eq. 2 of the PINK paper, } \delta = 1) \quad (4)$$

where  $\nu$  is Poisson’s ratio and  $\gamma$  the Grüneisen parameter — **both derived entirely from the predicted  $K$  and  $G$** , the same formula the paper’s own released `pink_predict.py` uses for real screening (its Table 1 validation numbers use a different, literature-looked-up  $\gamma$  instead; see `docs/method.tex`, Section 7.4, for why that distinction matters and is not a bug in this pipeline).

### Where the space-group data in this document comes from

Every Space Group / Space Group Number / Crystal System value below is read directly from GNoME’s own `stable_materials_summary.csv` — it is part of the structure metadata GNoME itself computed when the material was generated, joined onto our scored candidates by `MaterialId`. No new structure analysis was run for this document; 0.8–0.9% of candidates have no matching summary row (a small gap in the upstream file, not something this project introduces) and are reported, not silently dropped.

## 3 Crystallographic structure of the full screen

Table 2: Crystal system distribution, all 33,323 scored candidates (`scripts/cgcnn/17_spacegroups_general.py`). 264 candidates (0.8%) have no matching space-group row.

| Crystal system | Total candidates | Low- $\kappa_L$ candidates | % low- $\kappa_L$ |
|----------------|------------------|----------------------------|-------------------|
| Monoclinic     | 11,026           | 4,726                      | 42.9%             |
| Triclinic      | 10,761           | 6,802                      | 63.2%             |
| Orthorhombic   | 4,394            | 1,885                      | 42.9%             |
| Trigonal       | 4,151            | 1,632                      | 39.3%             |
| Tetragonal     | 1,352            | 575                        | 42.5%             |
| Hexagonal      | 749              | 298                        | 39.8%             |
| Cubic          | 626              | 254                        | 40.6%             |

Triclinic candidates clear the threshold at nearly  $1.5\times$  the rate of every other crystal system (63.2% vs. 39–43%) — the lowest-symmetry structures are disproportionately likely to be good  $\kappa_L$  candidates, which is physically sensible: low symmetry usually means a more complex, less uniform bonding environment, which scatters phonons more effectively.

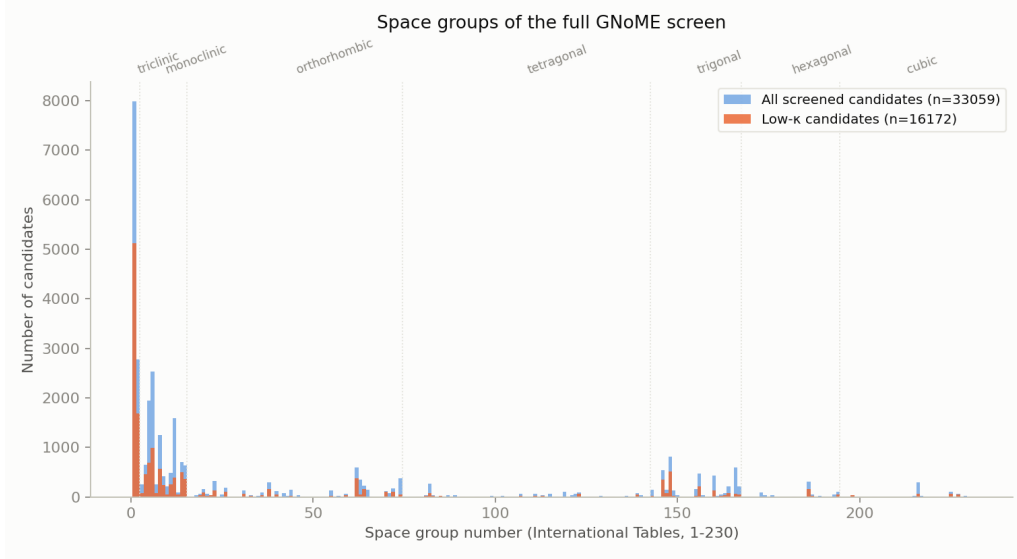


Figure 1: Space-group-number histogram, full screen. All candidates (blue) vs. the 16,299 low- $\kappa_L$  candidates (orange), with the seven crystal-system windows marked along the top axis.

### 3.1 Element composition and enrichment

For every element appearing anywhere in the screen, we compute how many candidates contain it and define an *enrichment ratio*: the fraction of low- $\kappa_L$  candidates containing that element, divided by its fraction of the whole screen. Enrichment  $> 1$  means that element’s compounds skew toward low  $\kappa_L$  more than average;  $< 1$  means the opposite.

Table 3: Most enriched and most depleted elements among low- $\kappa_L$  candidates (elements with  $\geq 30$  total candidates, to avoid noisy small counts; `scripts/cgcnn/18_element_frequency.py`).

| Element | Most enriched |              | Element | Most depleted |              |
|---------|---------------|--------------|---------|---------------|--------------|
|         | Total         | Enrichment   |         | Total         | Enrichment   |
| Br      | 3,319         | $1.88\times$ | Gd      | 776           | $0.17\times$ |
| I       | 3,221         | $1.86\times$ | O       | 11,612        | $0.27\times$ |
| Cl      | 1,378         | $1.78\times$ | W       | 1,357         | $0.29\times$ |
| Cs      | 3,628         | $1.73\times$ | Mo      | 1,809         | $0.35\times$ |
| Rb      | 3,863         | $1.66\times$ | B       | 858           | $0.37\times$ |
| C       | 711           | $1.64\times$ | N       | 1,239         | $0.40\times$ |
| F       | 2,579         | $1.62\times$ | Ca      | 1,455         | $0.42\times$ |
| K       | 3,144         | $1.50\times$ | Sr      | 2,280         | $0.50\times$ |
| Zr      | 1,240         | $1.43\times$ | Rh      | 998           | $0.56\times$ |
| Se      | 3,815         | $1.40\times$ | Mg      | 757           | $0.57\times$ |

### Halogens up, oxygen down

The four halogens present in appreciable numbers (Br, I, Cl, F) are all among the top seven most-enriched elements, alongside the large alkali cations they typically pair with (Cs, Rb, K). Oxygen, despite being the single most common element in the whole screen (11,612 of 33,323 candidates, 34.8%), is the second most *depleted* among low- $\kappa_L$  candidates at  $0.27\times$ . This is physically consistent: heavy, soft, highly polarizable halide lattices have lower sound velocities and Debye temperatures than the stiff, tightly-bonded lattices oxygen tends to form, and lower Debye temperature drives  $\kappa_L$  down directly through Eq. 3.

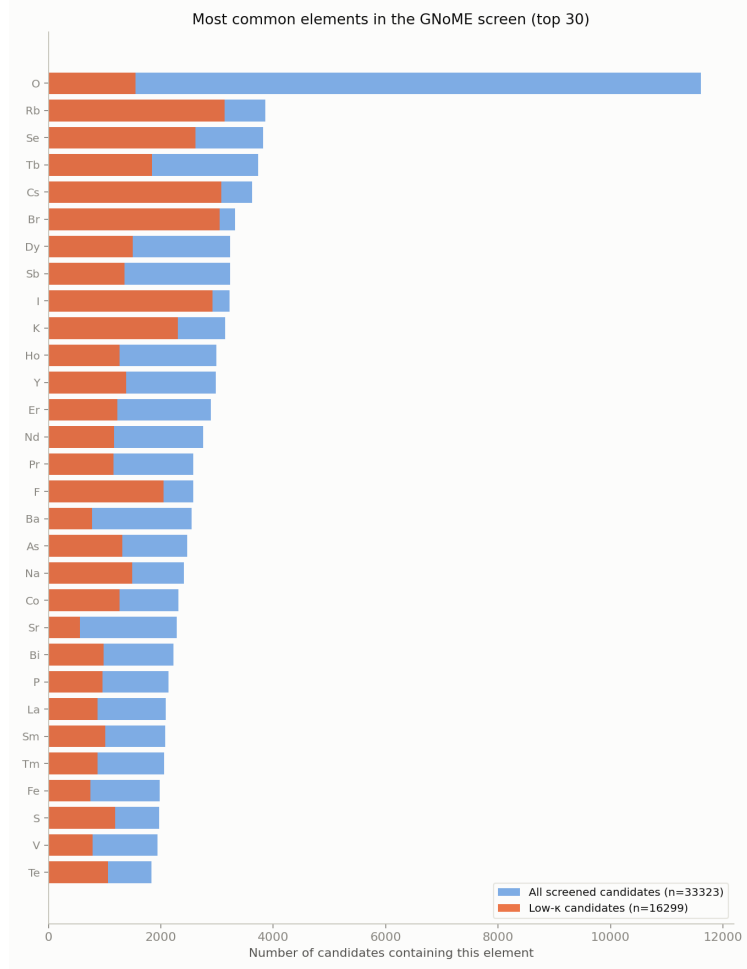


Figure 2: The 30 most common elements in the screen, all candidates (blue) vs. low- $\kappa_L$  candidates (orange).

## 4 The oxide subset

11,612 of the 33,323 candidates (34.8%) contain oxygen (`scripts/cgcnn/15_filter_oxides.py`), identified via pymatgen’s composition parser rather than a substring search (which would wrongly flag Os and Og). Only 1,545 of those 11,612 (13.3%) clear the low- $\kappa_L$  threshold — versus 48.9% for the screen as a whole — consistent with Section 3.1’s finding that oxygen is the second most depleted element.

Table 4: Crystal system distribution, oxide candidates only (`scripts/cgcnn/16_spacegroups_oxides.py`). 102 candidates (0.9%) have no matching space-group row.

| Crystal system | Total oxide candidates | Low- $\kappa_L$ candidates | % low- $\kappa_L$ |
|----------------|------------------------|----------------------------|-------------------|
| Monoclinic     | 4,280                  | 561                        | 13.1%             |
| Triclinic      | 3,373                  | 500                        | 14.8%             |
| Trigonal       | 1,747                  | 155                        | 8.9%              |
| Orthorhombic   | 1,415                  | 184                        | 13.0%             |
| Tetragonal     | 419                    | 94                         | 22.4%             |
| Hexagonal      | 176                    | 18                         | 10.2%             |
| Cubic          | 100                    | 22                         | 22.0%             |

Table 5: Stoichiometry pattern (reduced cation:oxygen ratio; `15_filter_oxides.py`’s own classification, a formula-ratio tag, not a DFT-verified structure). No exact ABO<sub>3</sub> (perovskite) or simple binary oxide survived the funnel — GNoME’s discovered materials skew toward complex, previously-uncatalogued compositions.

| Pattern                                                              | Total candidates | Low- $\kappa_L$ candidates |
|----------------------------------------------------------------------|------------------|----------------------------|
| Complex oxide (4 elements)                                           | 6,925            | 1,114                      |
| Complex oxide (5 elements)                                           | 4,110            | 230                        |
| Ternary oxide (A–B–O, other ratio)                                   | 548              | 185                        |
| A <sub>2</sub> B <sub>2</sub> O <sub>7</sub> -type (pyrochlore-like) | 21               | 2                          |
| AB <sub>2</sub> O <sub>4</sub> -type (spinel-like)                   | 8                | 3                          |

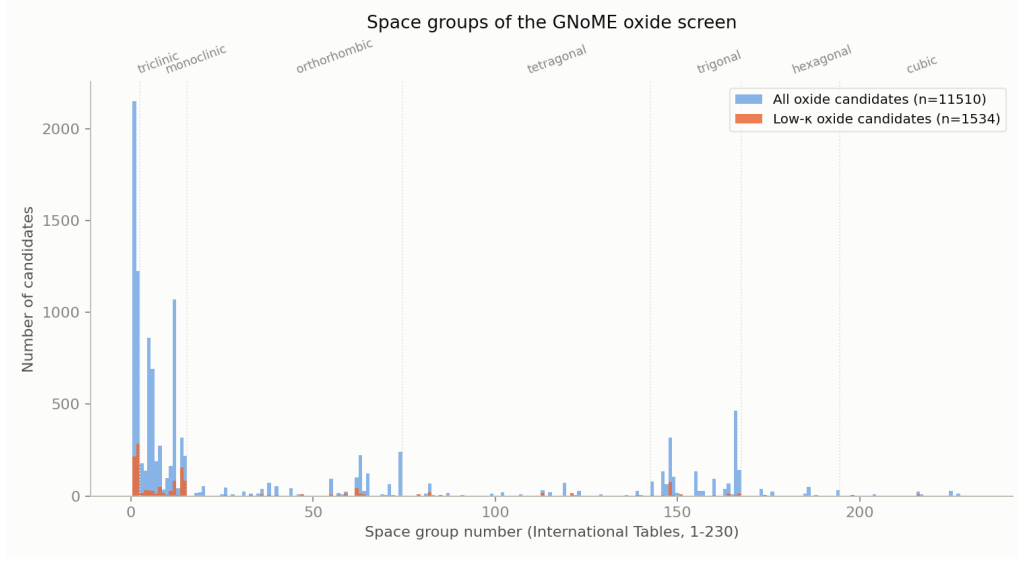


Figure 3: Space-group-number histogram, oxide candidates only. Same overlay/axis convention as Figure 1.

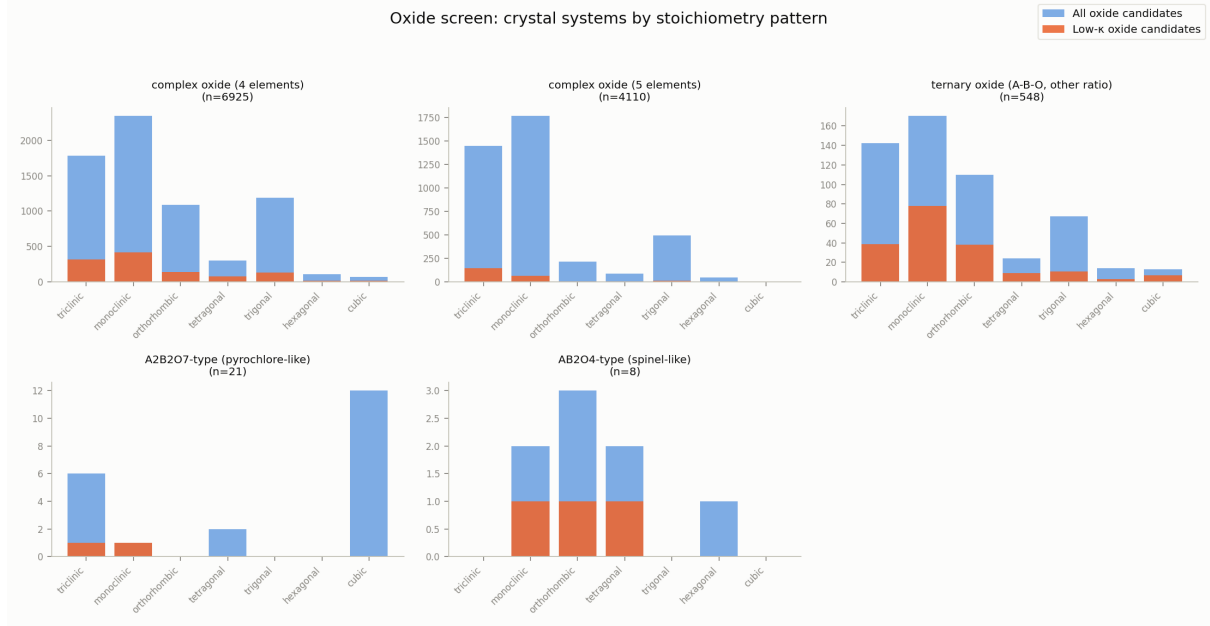


Figure 4: Crystal-system distribution within each stoichiometry pattern. Each panel has its own  $y$ -scale so the 8-row  $\text{AB}_2\text{O}_4$  pattern is not crushed flat next to the  $> 4,000$ -row complex-oxide patterns. The 21 pyrochlore-like candidates are notably cubic-dominated (12/21) — the canonical pyrochlore space group is cubic  $Fd\bar{3}m$  — but none of those 12 cubic pyrochlores clear the low- $\kappa_L$  threshold.

## 5 The strongest candidates

“Best” below means lowest  $\kappa_L$   $p95$  — the 95th-percentile Monte Carlo value from Section 2.2 — not the raw point estimate. Every row is shown regardless of whether it actually clears the  $\leq 1 \text{ W m}^{-1} \text{ K}^{-1}$  threshold, so a weak group is shown honestly rather than omitted.

### Why rank by p95, not the point estimate

Ranking 33,323 candidates by lowest point-estimate  $\kappa_L$  and taking the minimum is a text-book winner’s-curse setup: a candidate needs either a genuinely low  $\kappa_L$  or just a wide, noisy ensemble disagreement to land at the top, and the two look identical in a point estimate alone. Checked directly on this data: ranking by the point estimate instead of p95, the top oxide (ReAsOF<sub>10</sub>) and the second-best general candidate (HgWCl<sub>10</sub>) both carry p95/p05 ratios of 19–22× — solidly in the “worth a DFT check” range by this document’s own convention (Section 3), not the confident end of it. Ranking by p95 instead asks a more conservative question — *is this candidate still predicted low- $\kappa_L$  even in the model’s own pessimistic scenario* — and rewards confident, tightly-bounded predictions over lucky, noisy ones. The ratio column below makes that confidence visible rather than hiding it behind a single number.

Table 6: Best candidate per crystal system by  $\kappa_L$  p95, oxide vs. general screen. Ratio = p95/p05, this candidate’s own Monte Carlo spread — lower is more confident.

| Crys. sys.   | Formula | Oxide subset |                               | Formula        | Full screen |                        |
|--------------|---------|--------------|-------------------------------|----------------|-------------|------------------------|
|              |         | Sp. grp.     | p95 (ratio)                   |                | Sp. grp.    | p95 (ratio)            |
| Triclinic    | CuSCl6O | $P\bar{1}$   | 0.0756 (4.3×)                 | CrSi3Cl16      | $P1$        | <b>0.0094</b> (7.6×)   |
| Monoclinic   | COF14   | $P2_1$       | <b>0.0663</b> ( <b>2.3</b> ×) | TaSb(As2Cl13)2 | $Cm$        | 0.0191 ( <b>2.0</b> ×) |
| Orthorhombic | ReOF5   | $Pnma$       | 0.0636 (6.2×)                 | SbP2Cl13       | $Cmc2_1$    | 0.0125 (2.4×)          |
| Tetragonal   | AgN5O14 | $I\bar{4}$   | 0.1119 (7.5×)                 | POsCl10        | $P4/n$      | 0.0163 (4.2×)          |
| Trigonal     | S24ClO3 | $R3m$        | 0.0684 (3.8×)                 | FeSeBr18       | $R\bar{3}$  | 0.0264 (2.2×)          |
| Hexagonal    | Rb6BaO4 | $P6_3mc$     | 0.4867 (3.7×)                 | Rb3Co2Br9      | $P6_3/mmc$  | 0.1046 (4.8×)          |
| Cubic        | Fe3Br4O | $P2_13$      | 0.5684 (11.5×)                | CsRb2BiTe3     | $P2_13$     | 0.1079 (2.6×)          |

Bold marks each population’s single best candidate overall by this conservative ranking. **CrSi<sub>3</sub>Cl<sub>16</sub> holds its #1 spot even after re-ranking** — its 7.6× ratio, while not the tightest in the table, is still low enough that it stays ahead once every candidate’s own uncertainty is priced in, unlike HgWCl<sub>10</sub> (previously #2 by point estimate) which drops out of the top 10 entirely once ranked by p95. Among oxides, the point-estimate winner ReAsOF<sub>10</sub> is displaced by COF<sub>14</sub> — a much less dramatic  $\kappa_L$  but a 2.3× ratio, the tightest of any oxide crystal-system winner.

Table 7: Top 10 overall (one row per space group, ranked by  $\kappa_L$  p95), oxide subset. Full 134-row table: `gnome_oxide_best_by_spacegroup.csv` (also carries the point estimate and p05 for every row).

| #  | Formula       | Space group | Crystal system | p95    | Ratio        |
|----|---------------|-------------|----------------|--------|--------------|
| 1  | ReOF5         | $Pnma$      | orthorhombic   | 0.0636 | 6.2×         |
| 2  | COF14         | $P2_1$      | monoclinic     | 0.0663 | <b>2.3</b> × |
| 3  | S24ClO3       | $R3m$       | trigonal       | 0.0684 | 3.8×         |
| 4  | ReAsOF10      | $P2$        | monoclinic     | 0.0701 | 19.3×        |
| 5  | CeSCl6O       | $P2_1/c$    | monoclinic     | 0.0745 | <b>2.1</b> × |
| 6  | CuSCl6O       | $P\bar{1}$  | triclinic      | 0.0756 | 4.3×         |
| 7  | Rb2Re2OF10    | $P2/c$      | monoclinic     | 0.0837 | 3.9×         |
| 8  | TlRe2Ag(OF2)4 | $Cmm2$      | orthorhombic   | 0.0974 | 4.6×         |
| 9  | Na(S8O)3      | $P1$        | triclinic      | 0.1103 | 7.7×         |
| 10 | AgN5O14       | $I\bar{4}$  | tetragonal     | 0.1119 | 7.5×         |

Row 4 is the point-estimate winner from before ( $\kappa_L = 0.0158$ ) — still a strong candidate, just

no longer #1 once its own  $19.3\times$  uncertainty is weighed against tighter alternatives like row 2 or row 5.

Table 8: Top 10 overall (one row per space group, ranked by  $\kappa_L$  p95), full screen. Full 167-row table: `gnome_screen_best_by_spacegroup.csv`.

| #  | Formula        | Space group             | Crystal system | p95    | Ratio                         |
|----|----------------|-------------------------|----------------|--------|-------------------------------|
| 1  | CrSi3Cl16      | <i>P1</i>               | triclinic      | 0.0094 | $7.6\times$                   |
| 2  | SbP2Cl13       | <i>Cmc2<sub>1</sub></i> | orthorhombic   | 0.0125 | $2.4\times$                   |
| 3  | BI5            | <i>Pbcm</i>             | orthorhombic   | 0.0140 | $3.0\times$                   |
| 4  | BPI6           | <i>Pnma</i>             | orthorhombic   | 0.0161 | $4.1\times$                   |
| 5  | POsCl10        | <i>P4/n</i>             | tetragonal     | 0.0163 | $4.2\times$                   |
| 6  | TaSb(As2Cl13)2 | <i>Cm</i>               | monoclinic     | 0.0191 | <b><math>2.0\times</math></b> |
| 7  | YAg3Br28       | <i>Pm</i>               | monoclinic     | 0.0193 | $2.7\times$                   |
| 8  | Au3IBr12       | <i>P2<sub>1</sub></i>   | monoclinic     | 0.0210 | $4.5\times$                   |
| 9  | P(WCl6)2       | <i>P2<sub>1</sub>/m</i> | monoclinic     | 0.0212 | $4.8\times$                   |
| 10 | AgBr5          | <i>P2<sub>1</sub>/c</i> | monoclinic     | 0.0218 | $2.9\times$                   |

Every survivor here is again a heavy halide (Cl, Br), reinforcing Section 3.1’s enrichment finding independently of which ranking is used — the chemistry signal is robust to the point-estimate/p95 choice even though the individual winners are not.

## 6 Is the uncertainty interval actually calibrated?

Section 5’s entire premise — that ranking by  $\kappa_L$  p95 rather than the point estimate protects against noisy predictions — depends on one assumption never yet tested: that the ensemble’s claimed 90% interval (the gap between `Kappa_cal_p05` and `Kappa_cal_p95`) actually contains the true value about 90% of the time. `docs/method.tex` already flags this as “weak by construction,” since real experimental  $\kappa_L$  exists for only 2 materials our ensemble never trained on — nowhere near enough to check. But the uncertainty itself comes from disagreement on  $K$  and  $G$ , not  $\kappa_L$  directly, and the `matbench test` split — 1,648 crystals with real DFT-computed  $K$  and  $G$ , genuinely held out from training — already has everything needed to check that half, with no new data required.

### 6.1 The check

For each of the 1,648 test crystals, does the true  $\log_{10}$  modulus fall inside the ensemble’s own claimed interval ( $\text{pred} \pm z \cdot \sigma_{\text{ensemble}}$ ,  $z = 1.6449$  for a 90% normal interval — mathematically the same claim `monte_carlo_kappa()`’s 2,000-draw sampling converges to, just without sampling noise)? Repeated at every nominal level from 10% to 90%, not just the one used for screening.

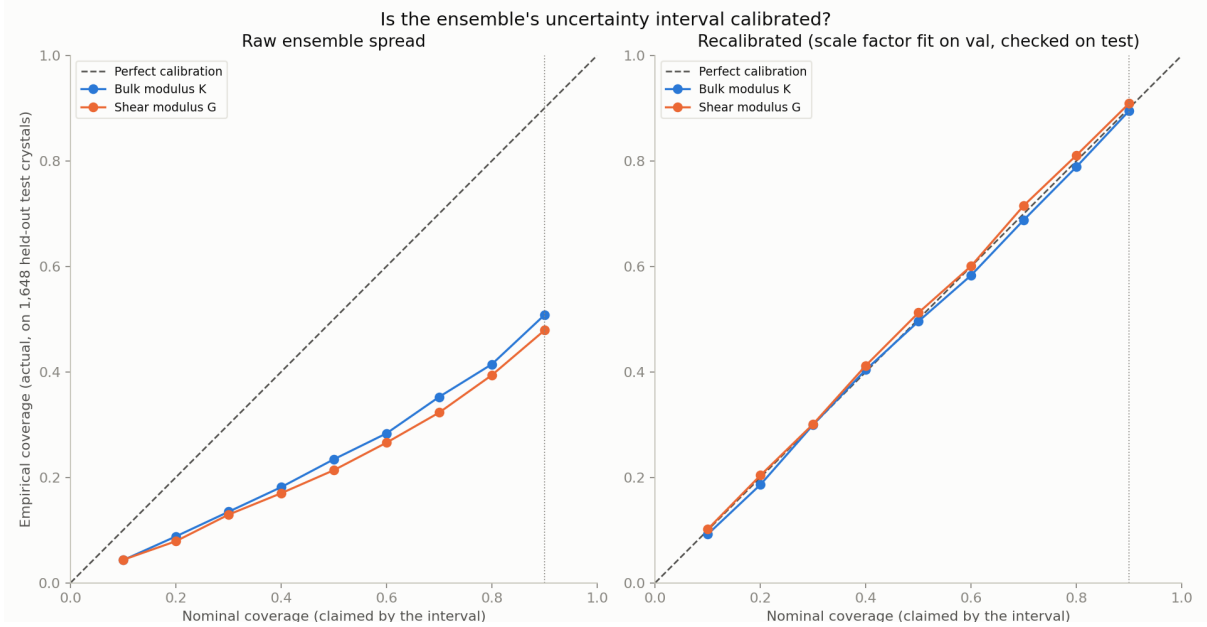


Figure 5: Reliability diagram: nominal coverage (claimed by the interval) vs. empirical coverage (actual, on 1,648 held-out test crystals). Left: the raw ensemble-spread interval used everywhere in this document up to this point. Right: after the recalibration in Section 6.2, with the correction factor fit on the (disjoint) validation split and checked on test.

### Badly overconfident, not roughly right

At the 90% level used throughout this document, the ACTUAL coverage is **50.8%** for  $K$  and **47.9%** for  $G$  — barely more than half of what the interval claims. The cause: ensemble spread measures how much three identically-trained models *disagree with each other*, which is a much smaller quantity than how far they collectively sit from the truth. Checked directly: median ensemble spread is 0.018–0.028  $\log_{10}$ , but the actual test RMSE is 0.118 for both moduli — the real error is 4–6 $\times$  larger than ensemble disagreement alone suggests. Three models sharing one architecture and one training set agree with each other even when they share the same blind spots.

## 6.2 The fix, checked out-of-sample

A scale factor derived from the *validation* split (a different 1,648 crystals from test) and applied to test — so the check is not circular — widens the interval to  $K$ : 4.27 $\times$ ,  $G$ : 4.54 $\times$  (on top of the untouched  $z = 1.645$ ), and recovers **89.5%** ( $K$ ) and **90.9%** ( $G$ ) actual coverage on test: right on target, out-of-sample.

## 6.3 What changes when this is applied to Section 5’s own picks

Nothing about the model changes —  $K_{\text{VRH}}$ ,  $G_{\text{VRH}}$ , and the point-estimate  $\kappa_L$  are byte-for-byte identical. Only how wide the error bars are reported changes. Re-scoring the same 14 crystal-system winners from Table 6 with the validated factor:

Table 9: Table 6’s own picks, p95 and ratio before vs. after recalibration. Every single one moves from “moderate”/“tight” into deep “wide” territory.

| Population | Formula          | Old p95 | Old ratio     | New p95      | New ratio       |
|------------|------------------|---------|---------------|--------------|-----------------|
| Oxide      | CuSCl6O          | 0.076   | 4.3 $\times$  | 0.734        | 417 $\times$    |
| Oxide      | <b>COF14</b>     | 0.066   | 2.3 $\times$  | <b>0.248</b> | 36 $\times$     |
| Oxide      | ReOF5            | 0.064   | 6.2 $\times$  | 0.908        | 1,972 $\times$  |
| Oxide      | AgN5O14          | 0.112   | 7.5 $\times$  | 1.956        | 2,591 $\times$  |
| Oxide      | S24ClO3          | 0.068   | 3.8 $\times$  | 0.404        | 273 $\times$    |
| Oxide      | Rb6BaO4          | 0.487   | 3.7 $\times$  | 3.070        | 193 $\times$    |
| Oxide      | Fe3Br4O          | 0.568   | 11.5 $\times$ | 15.090       | 14,474 $\times$ |
| General    | <b>CrSi3Cl16</b> | 0.009   | 7.6 $\times$  | 0.178        | 1,343 $\times$  |
| General    | <b>SbP2Cl13</b>  | 0.013   | 2.4 $\times$  | <b>0.042</b> | 31 $\times$     |
| General    | POsCl10          | 0.016   | 4.2 $\times$  | 0.121        | 250 $\times$    |
| General    | TaSb(As2Cl13)2   | 0.019   | 2.0 $\times$  | 0.050        | 16 $\times$     |
| General    | FeSeBr18         | 0.026   | 2.2 $\times$  | 0.099        | 34 $\times$     |
| General    | Rb3Co2Br9        | 0.105   | 4.8 $\times$  | 1.300        | 689 $\times$    |
| General    | CsRb2BiTe3       | 0.108   | 2.6 $\times$  | 0.411        | 40 $\times$     |

### One ranking survives intact, one does not

**COF14 (oxide)** holds its **#1 spot** even after recalibration — its new p95 (0.248) is still the lowest of the seven oxide picks, the same conclusion Section 5 reached with the uncalibrated interval. **CrSi3Cl16 (general)** **does not**: it was the overall #1 pick by the uncalibrated p95, but its own ratio explodes to  $1,343\times$  under the honest interval, and **SbP2Cl13** — previously ranked #2 in the space-group table (Table 8) — takes over as the new #1, with the tightest new ratio of the seven ( $31\times$ , still “wide” by the old cutoffs, but the best available). Every ratio in both populations inflates by roughly two to four orders of magnitude, a direct, honest consequence of a  $\log_{10}$ -space correction compounding once exponentiated back to linear  $\kappa_L$  — not a new error, the old numbers were simply wrong in the optimistic direction.

The fixed  $2\times/5\times$  tight/moderate/wide cutoffs (Section 3’s convention) were calibrated by eye against the *old*, artificially narrow interval; they do not carry over meaningfully once the interval is corrected, where every single one of these 14 headline picks now lands deep in “wide” territory. This is a known failure mode of small deep ensembles: disagreement among a handful of identically-architected, identically-trained members is a poor proxy for true predictive error, because it captures variance from random initialisation but not the bias every member shares. **The practical conclusion**: the *ranking* remains the most defensible thing to trust — COF14 and SbP2Cl13 are still the best *available* picks after recalibration — but no individual candidate’s uncertainty interval should be read as a genuine confidence guarantee until a fundamentally better-calibrated uncertainty method — a larger ensemble, conformal prediction, or an approach that captures model bias rather than only member disagreement — replaces the current one. This is now the single highest-leverage next step identified in this document, ahead of finding more candidates.

Reproduces via `scripts/cgcnn/25_calibration_check.py` (the check and the fix, on the matbench val/test splits) and `scripts/cgcnn/26_recalibrated_dft_shortlist.py` (the same correction applied to this project’s small-cell DFT shortlist, a further 267-candidate population screened alongside Sections 3–5 but not otherwise described in this document); recalibrated small-cell tables: `gnome_small_cell_dft_ranking_recalibrated.csv` and `gnome_small_cell_oxides_for_dft_recalibrated.csv`.

## A Appendix: CSV column reference

Every CSV referenced above shares the columns below. Columns already present in `gnome_screen_all.csv` (from `13_screen_gnome.py`) ride along unchanged; `space_group`, `space_group_number` and `crystal_system` are added by this document’s own scripts (`16_spacegroups_oxides.py`, `17_spacegroups_general.py`).

### Identity

| Column                              | Meaning                                                                        |
|-------------------------------------|--------------------------------------------------------------------------------|
| <code>material_id</code>            | GNoME’s own identifier for the material.                                       |
| <code>formula</code>                | Reduced chemical formula.                                                      |
| <code>stoichiometry_pattern</code>  | (oxide files only) reduced cation:oxygen ratio family — see Section 4’s table. |
| <code>is_low_kappa_candidate</code> | True if $\kappa_L \leq 1 \text{ W m}^{-1} \text{ K}^{-1}$ .                    |
| <code>space_group</code>            | / Read from GNoME’s own structure metadata — see the keybox in Section 2.2.    |
| <code>space_group_number</code>     |                                                                                |
| <code>/ crystal_system</code>       |                                                                                |

### Structural quantities (read from the primitive cell, not predicted)

|                               |                                                 |
|-------------------------------|-------------------------------------------------|
| Number of Atoms               | Atom count in the primitive cell.               |
| Volume (Å <sup>3</sup> )      | Primitive-cell volume, Å <sup>3</sup> .         |
| Density (g cm <sup>-3</sup> ) | Mass ÷ volume.                                  |
| Atomic mass (amu)             | Total composition weight of the primitive cell. |

### Model predictions (the 3-model CGCNN ensemble mean, linear GPa)

|                                 |                                                                                           |
|---------------------------------|-------------------------------------------------------------------------------------------|
| <code>K_VRH_pred</code>         | / Predicted bulk / shear modulus (GPa).                                                   |
| <code>G_VRH_pred</code>         |                                                                                           |
| <code>K_VRH_spread_log10</code> | / Ensemble disagreement, in $\log_{10}(\text{GPa})$ — a per-crystal uncertainty estimate. |
| <code>G_VRH_spread_log10</code> |                                                                                           |

### Derived physics (Slack model, Section 2.2) and Monte Carlo uncertainty

|                                                                    |                                                                                                                                                                                                                                                 |
|--------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Poisson ratio                                                      | / $\nu, \gamma$ above.                                                                                                                                                                                                                          |
| Gruneisen parameter                                                |                                                                                                                                                                                                                                                 |
| <code>Kappa_cal</code> (W m <sup>-1</sup> K <sup>-1</sup> )        | The predicted $\kappa_L$ — the number the whole screen thresholds on.                                                                                                                                                                           |
| <code>Kappa_cal_p05</code> / <code>_p50</code> / <code>_p95</code> | 5th/50th/95th percentile of $\kappa_L$ across 2,000 Monte Carlo draws of $K$ and $G$ (width set by the ensemble spread above). A tight interval means the ensemble agreed; a wide one means the candidate is worth a DFT check before trusting. |